08 Scientific Inquiry and Data Analysis Problem-solving Strategies

Basis: pattern definitions for INQ1–INQ7 in the CHE question-type catalogue `tools/catalogue_CHE.json`; evidence notes `evidence/che/notes/INQ.txt` (question stems, marking key lines and report notes for each sub-question); index `evidence/che/notes/INDEX.md` (marks and number of sub-questions). Most INQ questions do not depend on the data booklet; consult `CHE/papers/<年>_CHE_DataBooklet.pdf` only when solubility rules, standard electrode potentials or constants are needed. Question numbering: `2016S-Q26(a)` = 2016 Short answer (S), question 26, part (a); `E` = Extended answer; `2018E-Q35` has no part letter and is labelled as a whole question. Original questions and marking files are kept locally; this document only paraphrases the key's marking points and does not copy the original text. Review status: draft (catalogue status draft_for_review), not independently reviewed.

First, remember: where the marking panel repeatedly deducts marks

- Conclusions need evidence: explanation, prediction and evaluation questions (INQ5, INQ7) require every conclusion to cite a graph, table or calculation result. Writing only "increases" or "accurate" does not score; the key usually requires specific values, ranges or the location of a trend.
- Hypotheses must state the relationship and direction (INQ1): identifying only the variables and not writing their relationship scores at most half marks.
- Graph elements (INQ4): axis names and units, scale, plotted points, line of best fit; a title scores in 2021E-Q39(c) and 2022S-Q27(b), but the key for 2019S-Q34(a) explicitly gives no marks for a title. Missing any scored element loses marks.
- Safety measures must match specific hazards (INQ3): the key requires each measure to be linked to a specific hazard; a general "be careful" does not score.
- Errors must state direction (INQ6): the direction of random error is not fixed, while systematic error always deviates in one direction; reversing the two directions loses marks.
- ★ marking rule: patterns in the top quarter by INDEX marks are ★. There are 7 INQ patterns in total; INQ7 ranks first with 37 marks; INQ3, INQ5 and INQ6 tie for second (16 marks each). Taking the top quarter gives about 2 patterns, but the tie makes second place non-unique, so this document marks only INQ7.

Pattern 1: Hypotheses and inquiry questions

Identify: the question stem contains terms such as hypothesis, write an hypothesis, propose a hypothesis, construct a question, state a hypothesis, research question.

Steps

1. Identify the independent variable (the quantity the experimenter changes) and the dependent variable (the measured result); both must be measurable, and state the object and unit (e.g. temperature °C, yield %).
2. State the relationship between them: include the direction (increase or decrease, higher or lower) or trend (as ... increases, ...).
3. If the question asks for "based on information and prior knowledge", first state the reason (e.g. intermolecular forces, carbon chain length), then write a testable statement.
4. If a question form is required, the question should point to the relationship between the variables (e.g. "How does ... affect ...?").

Marking points (typical key wording):

- 2017E-Q39(g) (2 marks): a statement that gives both the relationship between the independent and dependent variables and the direction scores 2 marks; stating only that they are related without direction scores 1 mark.
- 2018S-Q33(a) (2 marks): a detailed statement based on the question stem information and relevant knowledge that can be tested experimentally scores 2 marks; a brief statement scores 1 mark.

- 2022E-Q39(a) (2 marks): stating the independent variable, dependent variable and their relationship scores 2 marks; listing only the two variables without a relationship scores 1 mark.
- 2021E-Q39(b) (1 mark): the question must show the relationship between temperature and biodiesel yield.

Common errors: listing variables without a relationship (at most half marks); writing the hypothesis as a fact or an untestable statement; writing only "affects" without direction; not using the information given in the question stem.

Past questions: 2017E-Q39(g), 2018S-Q33(a), 2021E-Q39(b), 2022E-Q39(a)

Pattern 2: Independent, dependent and controlled variables

Identify: the question stem contains identify the independent and dependent variables, state control variables, which trial(s) will allow ... to investigate the relationship, variables that need to be controlled.

Steps

1. Independent variable (IV): the quantity the experimenter deliberately changes; state the specific object and unit (e.g. "temperature of solution (°C)", "identity of the solvent").
2. Dependent variable (DV): the measured result; state the specific measurement (e.g. "time taken for ... to decolourise", "amount of paint removed").
3. Controlled variables: quantities other than the IV that could affect the result, such as temperature, volume, concentration, timing start point, apparatus; list two or more.
4. Choosing trials: identify trials in which only the target variable differs and all other conditions are the same (e.g. in 2021E-Q37(ai), the trial set in which only the sulfuric acid concentration is changed).

Marking points (typical key wording):

- Each correct variable scores 1 mark: 2020S-Q32(a) totals 2 marks (1 mark each for IV and DV); 2022E-Q39(c) totals 2 marks (1–2 marks) and requires different controlled variables to be listed.
- 2021E-Q37(ai) (1 mark): list all trial numbers that meet the condition.

Common errors: writing the DV too vaguely (e.g. "the reaction" or "colour"); confusing controlled variables with the IV, or writing the same quantity twice; in 2021E-Q37(ai), listing only some trial groups and missing other groups with the same conditions.

Past questions: 2020S-Q32(a), 2020S-Q32(b), 2021E-Q37(ai), 2022E-Q39(b), 2022E-Q39(c)

Pattern 3: Experimental design, procedure and safety

Identify: the question stem contains describe a procedure, outline the procedure, safety measures, personal safety, health risk, hazard, minimise the risk, suggest ... safety measures.

Steps

- Experimental procedure:
- Fixed quantities: state the volume or mass of each reagent and the apparatus used (e.g. pipette, volumetric flask).
- Variable handling: change the independent variable and measure its value (e.g. temperature must be varied and recorded).
- Measurement method: state the timing start point and endpoint (e.g. "start timing from mixing") and the method for judging the endpoint (e.g. white paper underneath to judge colour loss).
- Repeats and ratios: state repeat trials and the mixing ratio of solutions.
- Safety:
- State the specific hazard (e.g. corrosive alkali, cyanide or the HCN gas it releases, dust, vapour).
- State the corresponding measure (e.g. safety goggles, gloves, lab coat, sealed container, ventilation, checking the material safety data sheet).

- Each measure must be linked to a specific hazard.

Marking points (typical key wording):

- 2022E-Q39(d) (6 marks): fixed volumes of the two reagents (1–2 marks); temperature changed and measured (1 mark); timing from mixing (1 mark); any two of endpoint method, repeat trials, apparatus, mixing ratio (2 marks).
- 2018S-Q26(d) (2 marks): list two personal protective measures.
- 2019S-Q31(d) (3 marks): each measure linked to a specific hazard scores 1 mark (the key hazards are the toxicity of cyanide ions and the production of HCN gas).
- 2020S-Q32(d) (4 marks): 1 mark for each of two risks, and 1 mark for the method of reducing each risk.
- 2017E-Q39(f) (1 mark): state one health hazard of NaOH (e.g. burns, irritation).

Common errors: general descriptions (e.g. "be careful") do not score; stating measures without hazards (2019S-Q31(d)); omitting the timing start point in the procedure (2022E-Q39(d)); naming apparatus only without explaining how to measure; writing a procedure that cannot be repeated (missing volumes or ratios).
Past questions: 2017E-Q39(f), 2018S-Q26(d), 2019S-Q31(d), 2020S-Q32(d), 2022E-Q39(d)

Pattern 4: Drawing graphs

Identify: the question stem contains draw a labelled graph, graph the data, plot, use the grid, spare grid, graph the ... versus ....
Steps

1. Title: state the relationship between the dependent and independent variables (e.g. "yield changes with temperature"); only needed when the question awards marks for a title (e.g. 2021E-Q39(c)); 2019S-Q34(a) does not award marks for it.
2. Put the independent variable on the horizontal axis and the dependent variable on the vertical axis; label names and units (e.g. °C, %).
3. Scale: uniform and suitable, covering the data range, not compressed into one corner of the graph, and no broken scale.
4. Plotted points: place each data point accurately.
5. Draw a line or curve of best fit; do not join the points with straight line segments.
6. If using a spare graph, cross out the original and note that it has been redrawn on the spare page (as required by the question).

Marking points (typical key wording):

- 2019S-Q34(a) (4 marks): axis names and units (1 mark), scale (1 mark), plotted points (1 mark), line of best fit (1 mark).
- 2021E-Q39(c) (5 marks): title (1 mark), scale (1 mark), axis names and units (1 mark), plotted points (1 mark), line of best fit (1 mark).
- 2022S-Q27(b) (5 marks): 1 mark each for title, vertical axis unit, scale, plotted points, line of best fit.

Common errors: omitting the vertical axis unit (e.g. %) or temperature unit (°C); a title that does not state the variable relationship; joining data points with straight line segments; relabelling an axis that is already labelled; forgetting to cross out the original graph when using a spare graph.
Past questions: 2019S-Q34(a), 2021E-Q39(c), 2022S-Q27(b)

Pattern 5: Interpreting data and trends, extrapolation

Identify: The question stem contains describe the relationship, use your graph to predict, justify your prediction, show how the tabulated data ... can be used to deduce, calculations are not required, what is the reading.

Steps

1. Reading: read to the precision required by the instrument or the question (e.g. voltage to two decimal places).
2. Describe the trend: state the direction, range and special points (e.g. it first rises, reaches a maximum at a certain temperature, then falls).
3. Extrapolate or interpolate: read values along the line of best fit, and state that it was read from the graph.
4. Chain of evidence: data → intermediate reasoning → conclusion. For example, pH decreases → $[H^+]$ increases → the forward reaction is favoured → by Le Châtelier's principle, deduce that it is an endothermic reaction.
5. When the question asks to "justify", each step must cite specific evidence from the graph or table.

Marking points (typical wording in the key):

- 2021S-Q29(a) (5 marks): the trend in the pH data, $[H^+]$ increasing, the forward reaction being favoured, deducing the endothermic direction by Le Châtelier's principle, and the conclusion, one step each. The question states that calculations are not required, so the answer should focus on the qualitative chain of reasoning.
- 2020S-Q29(c) (4 marks): K decreases (1 mark); in the graph $n(H_2)$ decreases as temperature increases (1 mark); any two pieces of evidence for equilibrium shift (2 marks).
- 2021E-Q37(b) (3 marks): conclusion "yes" (1 mark); the trial with Fe^{2+} dissolves more Co (1 mark); Fe^{2+} itself is not consumed (1 mark).
- 2021E-Q39(d) (2 marks): state that the yield first increases and reaches a maximum at 55 °C (1 mark); the yield decreases at higher temperatures (1 mark).
- 2022S-Q27(c) (1 mark): read 104 °C from the line of best fit.

Common mistakes: only saying "increases" without stating the position of the maximum (2021E-Q39(d)); not citing graph data when predicting (2020S-Q29(c) requires justify); performing calculations when the question prohibits them; ignoring the shape of the curve when extrapolating and using a linear relationship directly.

Past exam questions: 2017S-Q34(ci), 2020S-Q29(c), 2021S-Q29(a), 2021E-Q37(b), 2021E-Q39(d), 2022S-Q27(c)

Pattern 6: Errors, accuracy, reliability and sampling

Identify: The question stem contains random and systematic errors, outline the difference, give an example, sample size, as a percentage of, reasons why samples were not collected, reliable, valid, systematic, consistently.

Steps

1. Random error: unpredictable fluctuations; readings may be too high or too low, with no fixed direction; can be reduced by repeating trials and averaging.
2. Systematic error: recurring deviation from the true value in the same direction (e.g. an uncalibrated instrument, the labelled concentration not matching the actual concentration, a bias in the reading method); requires correcting the instrument or improving the design.
3. Examples must come from the specific steps of this experiment, e.g. "judging the endpoint" is a random error, while "the acid concentration not matching the label" is a systematic error.
4. Sampling proportion = number of samples ÷ number of possible sampling points × 100% (e.g. 59 ÷ 180 × 100 = 32.8%). A small sample is less representative, reliability decreases, and the effect of random error increases.
5. The four concepts must be kept separate (the following are the author's definitions; the key mainly uses valid and reliable): accuracy means the result is close to the true value; precision means repeated measurements are close to each other; validity (valid) means the experimental design can answer the question (controlling variables, comparing the correct things). Reliability (reliable) means repeated

trials under the same conditions give consistent results.

Marking points (typical wording in the key):

- 2018E-Q36(a) (4 marks): definition of random error and systematic error, 1 mark each; one experiment-related example of each, 1 mark each.
- 2022E-Q39(e) (4 marks): definition and example of systematic error; definition and example of random error.
- 2018E-Q36(bi) (1 mark): calculate 32.8%.
- 2018E-Q36(bii) (2 marks): state any two reasons why all locations were not sampled (high cost, time-consuming, difficult to access, the sample is statistically sufficient).
- 2018E-Q36(biii) (2 marks): predict that reliability decreases, and give a reason (few sampling locations, difficult to cover the full concentration range, greater effect of random error).
- 2019S-Q35(c) (1 mark): the error always makes readings too high or too low.
- 2020S-Q32(ci) (1 mark): validity requires comparing the effect of different solvents on paint removal, and controlling variables.
- 2020S-Q32(cii) (1 mark): explain reliability by obtaining consistent results through repeated trials.

Common mistakes: reversing the direction of random and systematic error (random error has no fixed direction); giving examples unrelated to this experiment, such as a vague "human error"; using the wrong denominator for the sampling proportion (the denominator is the total number of possible sampling points, 180, not the 59 sampled, then multiply by 100%); writing "reliable" as "lots of data" without mentioning repetition; confusing accuracy (close to the true value) with reliability (consistent on repetition).

Past exam questions: 2018E-Q36(a), 2018E-Q36(bi), 2018E-Q36(bii), 2018E-Q36(biii), 2019S-Q35(c), 2020S-Q32(ci), 2020S-Q32(cii), 2022E-Q39(e)

Pattern 7: Evaluating conclusions and claims ★

Identify: The question stem contains demonstrate whether or not, comment on the accuracy, justify your answer, determine whether the claim is true, use a calculation to determine, predict the identity, determine if it would be classified as, does the ... support.

Steps

1. Identify the claim in the question: label value, claimed efficiency, dosage in the instructions, a promise such as "at least x%".
2. Use the calculations from the previous patterns to find the experimental value (titration, yield, empirical formula, potential difference, etc.).
3. Determine the acceptable range: e.g. with a permitted error of ±5%, centred on the label value 295 g L^{-1}, the range is 280.25–309.75 g L^{-1} (2019E-Q39(c)).
4. Compare and give a clear conclusion (e.g. "does not match the label", "the claim is not true", "the concentration is accurate").
5. Give justification: list the two values used for comparison. If the question requires evaluating data quality, add the source and direction of the error.
6. Prediction type (e.g. 2017S-Q34(cii)): calculate the potential difference range of the other electrode from the voltage, compare with the standard electrode potential table to select possible metals, then use qualitative evidence (e.g. forming a colourless ion or a +2 ion) to determine the unique answer.

Marking points (typical wording in the key):

- 2019E-Q39(c) (8 marks): multi-step calculation to obtain the experimental concentration (about 276 g L^{-1}), convert to g L^{-1}, give the acceptable range, and conclude "does not match".
- 2022S-Q31(b) (3 marks): calculate the required water absorption (715 g), correct by 97.4% (696 g), compare with 691 g and conclude "the claim is not true".
- 2024E-Q36(c) (8 marks): calculate the mass per tablet (305 mg), compare with the label's "double

dose", conclude that the tablet is actually the standard dose.

- 2025E-Q38(d) (7 marks): each titration value 1 mark, average 1 mark, n(OH^-) 1 mark, stoichiometric ratio 1 mark, mass and concentration 2 marks, conclusion "the label concentration is accurate" 1 mark.
- 2017S-Q34(cii) (3 marks): calculation and range of potential difference (1 mark), basis for selection from the table (1 mark), conclusion Zn (1 mark).
- 2016E-Q38(d) (2 marks): compare the calculated empirical formula with the target product (choline), confirming the enzyme-catalysed product rather than the substrate.
- 2016E-Q42(f) (6 marks): pH calculation and acid rain criterion. The criterion given in the question is that acid rain requires pH below 4.4; pH 5.20 is not acid rain; no calculation process means no marks.

Common mistakes:

- Conclusion inconsistent with the numbers: the experimental value in 2019E-Q39(c) is 276.3 g L^{-1}, which lies outside the acceptable range, so the conclusion must be "does not match".
- Not using the criterion given in the question: for 2016E-Q42(f), the acid rain criterion must be based on the question, not applied from other sources.
- Known issue with the official key (2024E-Q36(c)): the main key calculates using OH^- : aspirin = 1 : 1, giving 305 mg/tablet; the key note says that if a 2 : 1 ratio is used, it gives 152 mg. The two results differ by a factor of two, and the key is internally inconsistent (see the note in QNT4). Students should state the stoichiometric ratio used and its basis, and make the conclusion about the label accordingly.
- Different comparison bases: the two numbers in 2022S-Q31(b) must both be "water absorption mass"; one cannot be a mass and the other a percentage.

Past exam questions: 2016E-Q38(d), 2016E-Q42(f), 2017S-Q34(cii), 2019E-Q39(c), 2022S-Q31(b), 2024E-Q36(c), 2025E-Q38(d)